

Data Analytics (DA) Module -2

Power BI Assignment 2 – DAX & Data Visualization

Dataset Description :

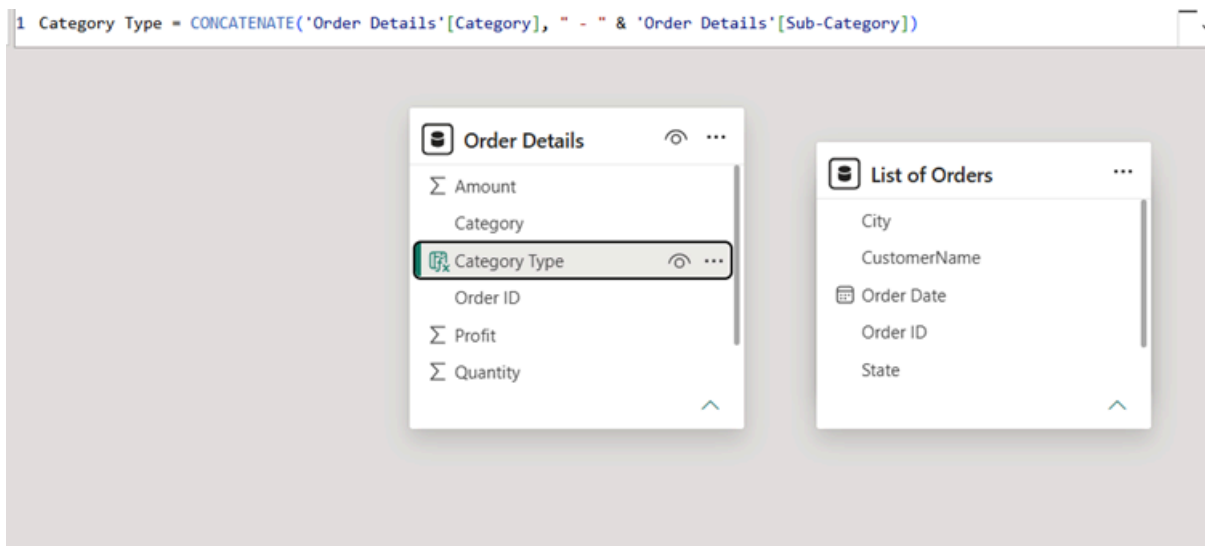
You will work with the following three datasets:

- List of orders Attributes: Order ID, Order Date, Customer Name, City, State, Category, Amount, Profit
- Order Details Attributes: Order ID, Category, Sub-Category, Quantity, Amount, Profit
- Sales Target Attributes: Category, Month of Order Date, Target

Task

Calculated Columns

1. Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type
B-25863	189	87	7	Clothing	Stole	Clothing - Stole
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree
B-25604	157	5	9	Clothing	Saree	Clothing - Saree
B-25605	75	0	7	Clothing	Saree	Clothing - Saree
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree
B-25610	43	0	3	Clothing	Saree	Clothing - Saree
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree
B-25619	353	90	8	Clothing	Saree	Clothing - Saree
B-25622	534	0	3	Clothing	Saree	Clothing - Saree
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree
B-25628	24	-9	4	Clothing	Saree	Clothing - Saree
B-25633	711	-8	4	Clothing	Saree	Clothing - Saree
B-25635	382	30	3	Clothing	Saree	Clothing - Saree
B-25636	637	113	5	Clothing	Saree	Clothing - Saree
B-25640	122	-47	4	Clothing	Saree	Clothing - Saree
B-25646	20	-8	2	Clothing	Saree	Clothing - Saree
B-25647	42	-6	4	Clothing	Saree	Clothing - Saree
B-25648	55	-26	4	Clothing	Saree	Clothing - Saree
B-25648	130	-41	4	Clothing	Saree	Clothing - Saree
B-25650	211	-105	2	Clothing	Saree	Clothing - Saree
B-25650	31	-2	2	Clothing	Saree	Clothing - Saree
B-25650	512	-225	5	Clothing	Saree	Clothing - Saree
B-25650	238	20	2	Clothing	Saree	Clothing - Saree
B-25651	23	4	1	Clothing	Saree	Clothing - Saree
B-25651	457	41	4	Clothing	Saree	Clothing - Saree

2.Calculate Revenue per Order in Order Details Table:

Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

1 Revenue per Order = 'Order Details'[Amount] * 'Order Details'[Quantity]

Order Details
Σ Amount
Category
Category Type
Order ID
Σ Profit
Σ Quantity

List of Orders
City
CustomerName
Order Date
Order ID
State

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order
B-25863	189	87	7	Clothing	Stole	Clothing - Stole	1323
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	952
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	579
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	1413
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	525
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	100
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	129
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	320
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	14427
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	2824
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	1602
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	596
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	3175
B-25628	24	-9	4	Clothing	Saree	Clothing - Saree	96
B-25633	711	-8	4	Clothing	Saree	Clothing - Saree	2844
B-25635	382	30	3	Clothing	Saree	Clothing - Saree	1146
B-25636	637	113	5	Clothing	Saree	Clothing - Saree	3185
B-25640	122	-47	4	Clothing	Saree	Clothing - Saree	488
B-25646	20	-8	2	Clothing	Saree	Clothing - Saree	40
B-25647	42	-6	4	Clothing	Saree	Clothing - Saree	168
B-25648	55	-26	4	Clothing	Saree	Clothing - Saree	220
B-25648	130	-41	4	Clothing	Saree	Clothing - Saree	520
B-25650	211	-105	2	Clothing	Saree	Clothing - Saree	422
B-25650	31	-2	2	Clothing	Saree	Clothing - Saree	62
B-25650	512	-225	5	Clothing	Saree	Clothing - Saree	2560
B-25650	238	20	2	Clothing	Saree	Clothing - Saree	476
B-25651	23	4	1	Clothing	Saree	Clothing - Saree	23
B-25651	457	41	4	Clothing	Saree	Clothing - Saree	1828

Order Details (1,500 rows) Column: Revenue per Order (845 distinct values)

3.Create a Calculated Column to Categorize Sales: Add a calculated column named 'Sales Category' in the Order Details table that

categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

Sales Category =
IF(
 'Order Details'[Amount] > CALCULATE(AVERAGE('Order Details'[Amount]), ALL('Order Details')),
 "Above Average",
 "Below Average"
)

Σ Amount

Category
Category Type
Order ID
Σ Profit
Σ Quantity
Revenue per Order
Sales Category
Sub-Category

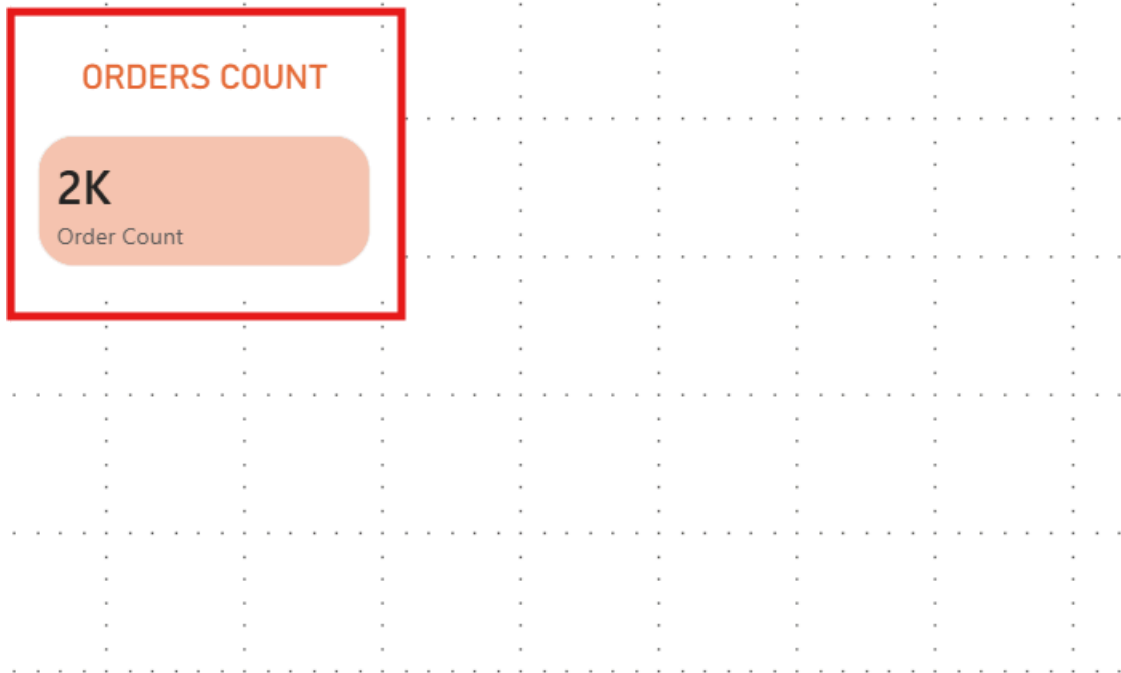
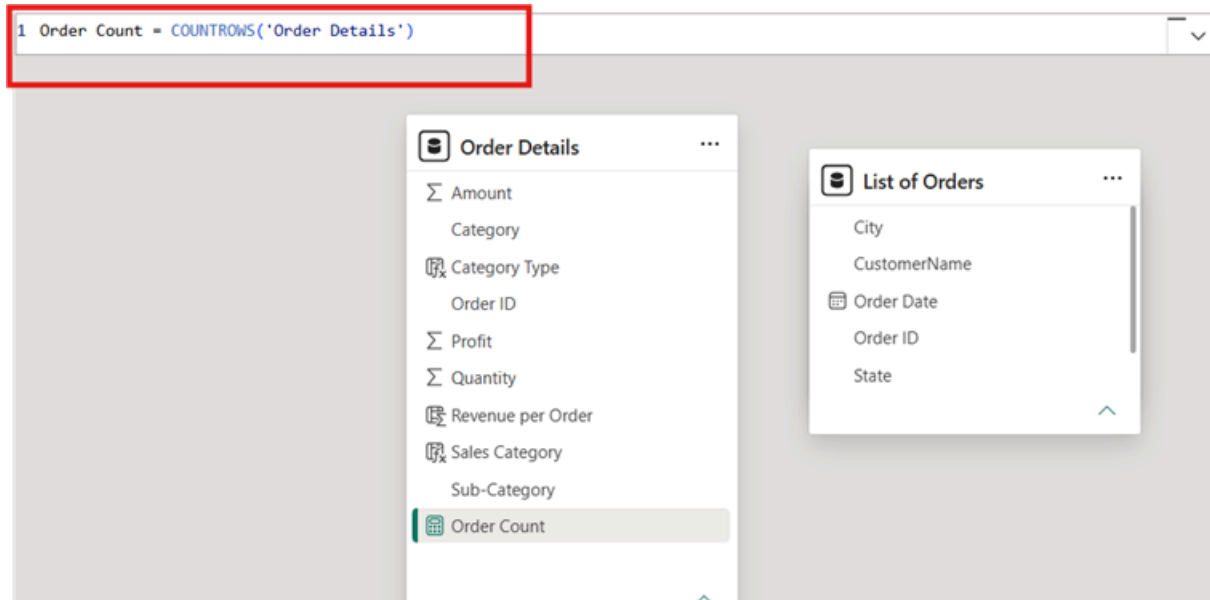
List of Orders
City
CustomerName
Order Date
Order ID
State

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order	Sales Category
B-25863	189	87	7	Clothing	Stole	Clothing - Stole	1323	Below Average
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	952	Below Average
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	1413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	529	Below Average
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	14427	Above Average
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	2824	Above Average
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	1602	Above Average
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	596	Below Average
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	3175	Above Average
B-25628	24	-9	4	Clothing	Saree	Clothing - Saree	96	Below Average
B-25633	711	-8	4	Clothing	Saree	Clothing - Saree	2844	Above Average
B-25635	382	30	3	Clothing	Saree	Clothing - Saree	1146	Above Average
B-25636	637	113	5	Clothing	Saree	Clothing - Saree	3185	Above Average
B-25640	122	-47	4	Clothing	Saree	Clothing - Saree	488	Below Average
B-25646	20	-8	2	Clothing	Saree	Clothing - Saree	40	Below Average
B-25647	42	-6	4	Clothing	Saree	Clothing - Saree	168	Below Average
B-25648	55	-26	4	Clothing	Saree	Clothing - Saree	220	Below Average
B-25648	130	-41	4	Clothing	Saree	Clothing - Saree	520	Below Average
B-25650	211	-105	2	Clothing	Saree	Clothing - Saree	422	Below Average
B-25650	31	-2	2	Clothing	Saree	Clothing - Saree	62	Below Average
B-25650	512	-225	5	Clothing	Saree	Clothing - Saree	2560	Above Average
B-25650	238	20	2	Clothing	Saree	Clothing - Saree	476	Below Average
B-25651	23	4	1	Clothing	Saree	Clothing - Saree	23	Below Average
B-25651	157	11	4	Clothing	Saree	Clothing - Saree	1028	Above Average

Order Details (1,500 rows) Column: Sales Category (2 distinct values)

Calculated Measures

1. Order Count- Create a measure that calculates the total number of orders in the Order Details table.



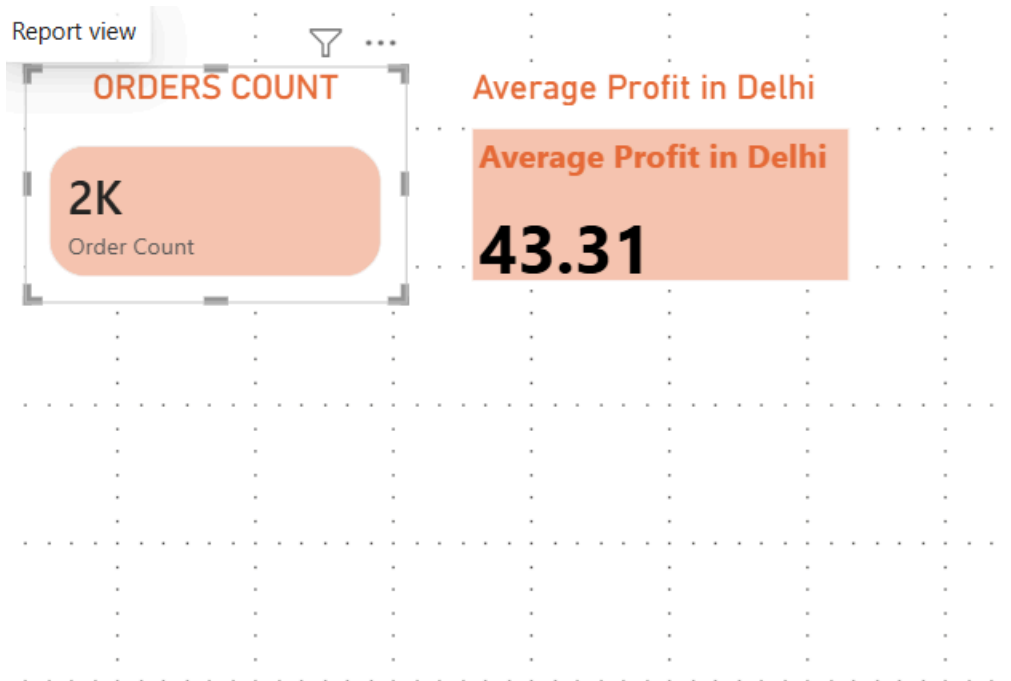
2.Average Profit in Delhi- Create a measure that calculates the average profit for orders placed in Delhi.

```
1 Average Profit in Delhi =
2 CALCULATE(
3     AVERAGE('Order Details'[Profit]),
4     'List of Orders'[City] = "Delhi"
5 )
```

The screenshot shows the DAX editor with the following formula:

```
Average Profit in Delhi =
CALCULATE(
    AVERAGE('Order Details'[Profit]),
    'List of Orders'[City] = "Delhi"
)
```

The data model shows two tables: 'Order Details' and 'List of Orders'. The 'List of Orders' table has columns: City, CustomerName, Order Date, Order ID, and State. The 'Order Details' table has columns: Amount, Category, Category Type, Order ID, Profit, Quantity, Revenue per Order, Sales Category, Sub-Category, and Order Count. A relationship is shown between the 'Order ID' columns of both tables.



3. Year-to-Date (YTD) Sales- Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.

1 YTD Sales =
2 TOTALYTD(
3 SUM('Order Details'[Amount]),
4 'List of Orders'[Order Date]
5)

Order Details

Σ Amount

Category

Category Type

Order ID

Σ Profit

Σ Quantity

Revenue per Order

Sales Category

Sub-Category

Average Profit in Delhi

Order Count

List of Orders

City

CustomerName

Order Date

Order ID

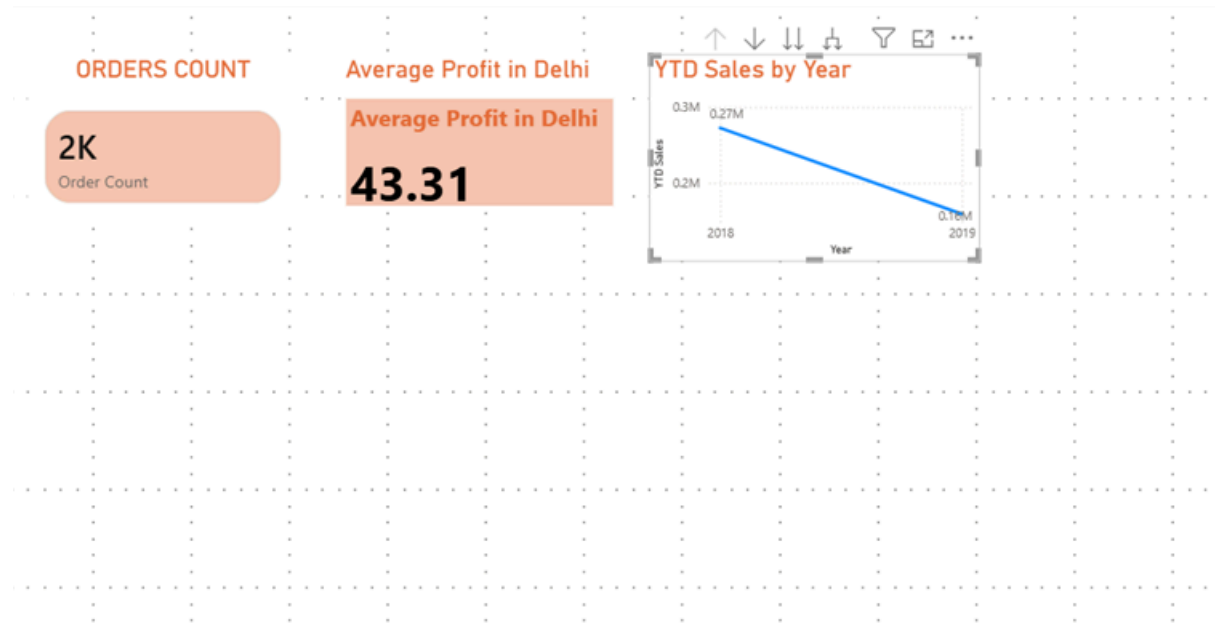
State

Category Type

YTD Sales

*

1



Data Visualization

Create the following visualizations using Power BI:

1. **Sales Target Achievement by Category:** Compare actual sales with sales targets by category using a clustered column chart.


```
1 Sales Target = SUM('Sales Target'[Target])
```

Order Details

Σ Amount
Category
Category Type
Order ID
Σ Profit
Σ Quantity
Revenue per Order
Sales Category
Sub-Category
Average Profit in Delhi
Order Count

List of Orders

City
CustomerName
Order Date
Order ID
State
Category Type
YTD Sales

Sales target

Category
Month of Order Date
Σ Target
Sales Target

ORDERS COUNT

2K

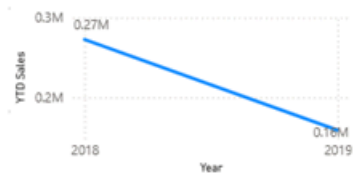
Order Count

Average Profit in Delhi

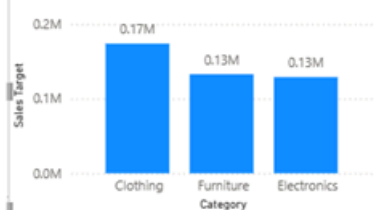
Average Profit in Delhi

43.31

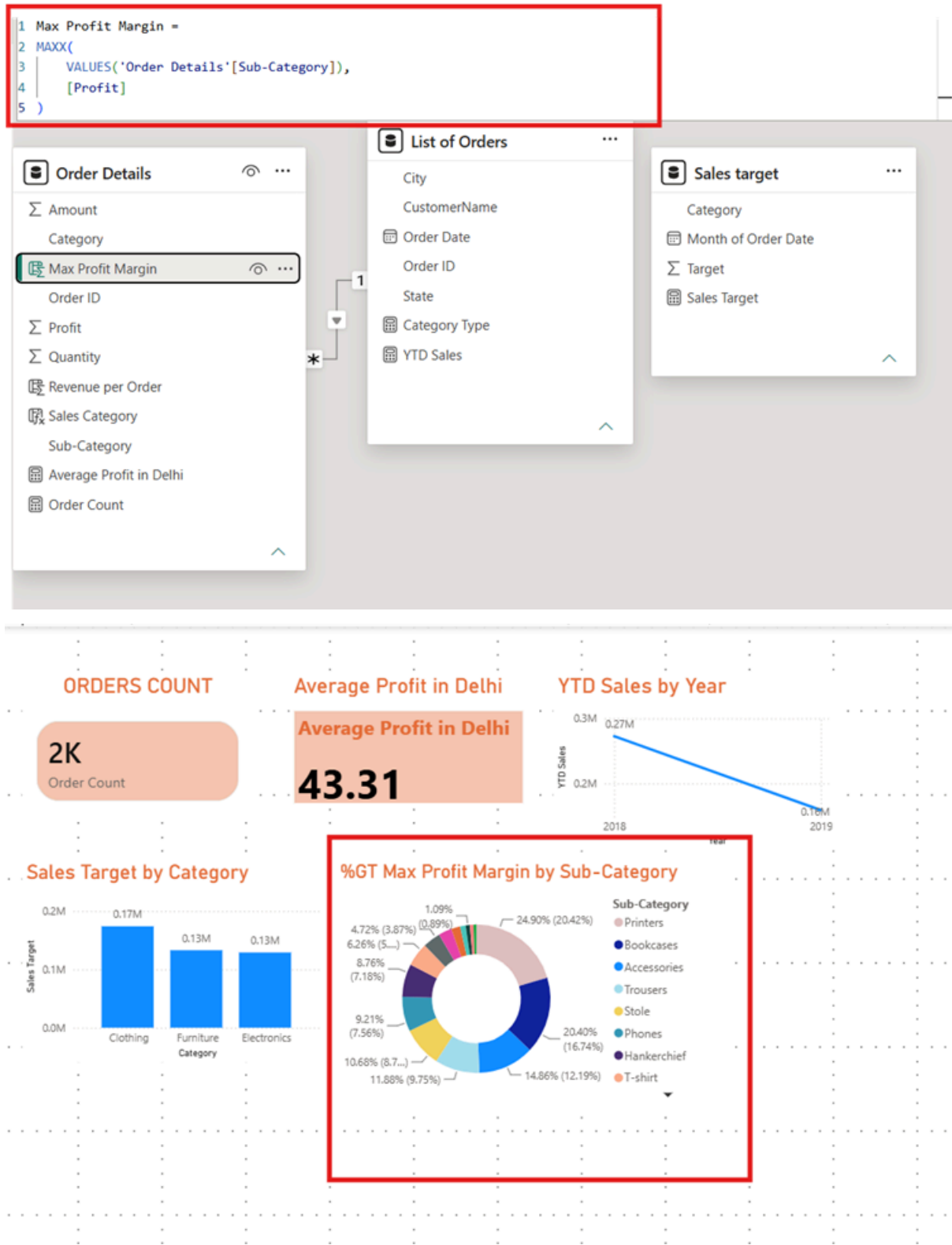
YTD Sales by Year



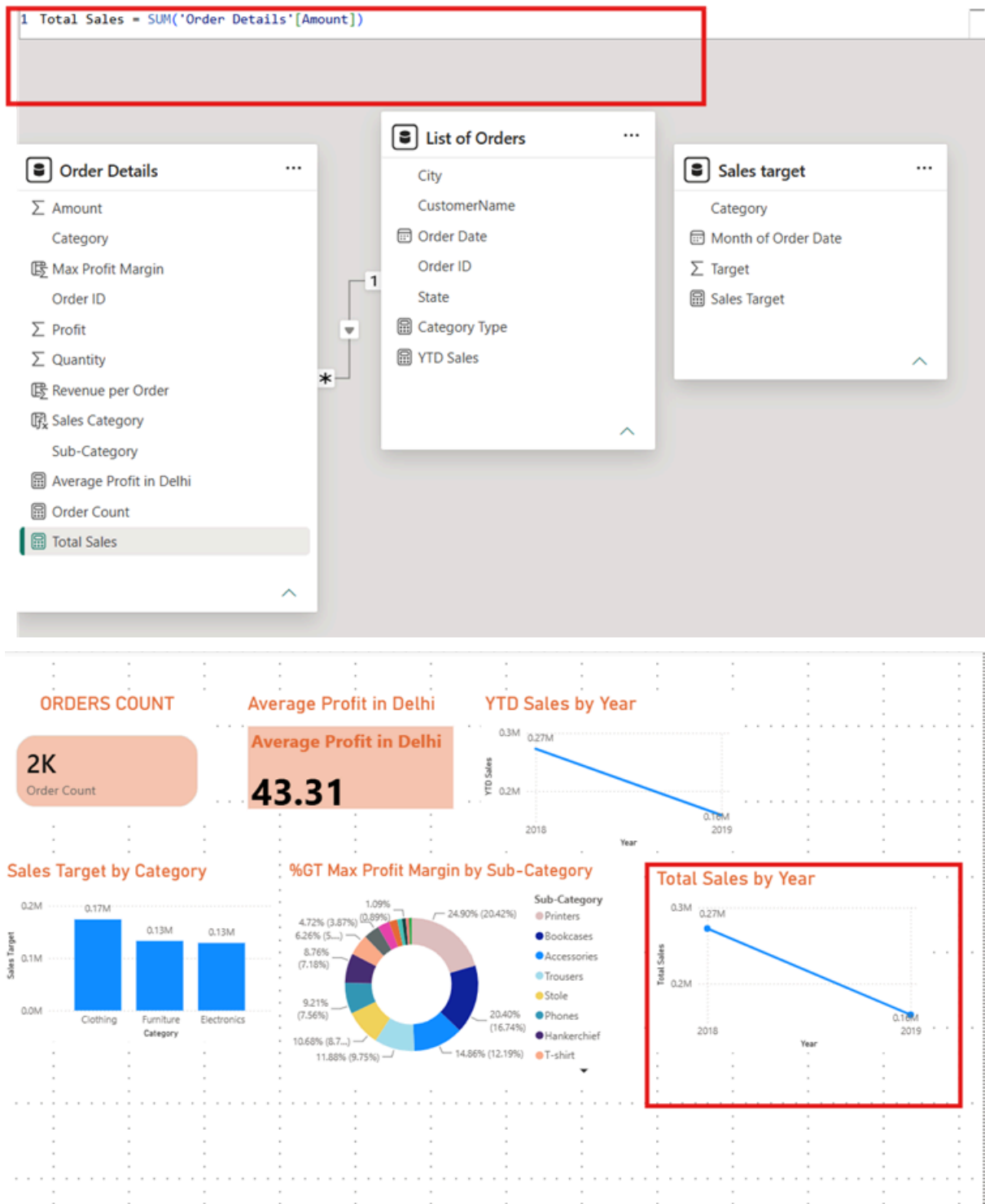
Sales Target by Category



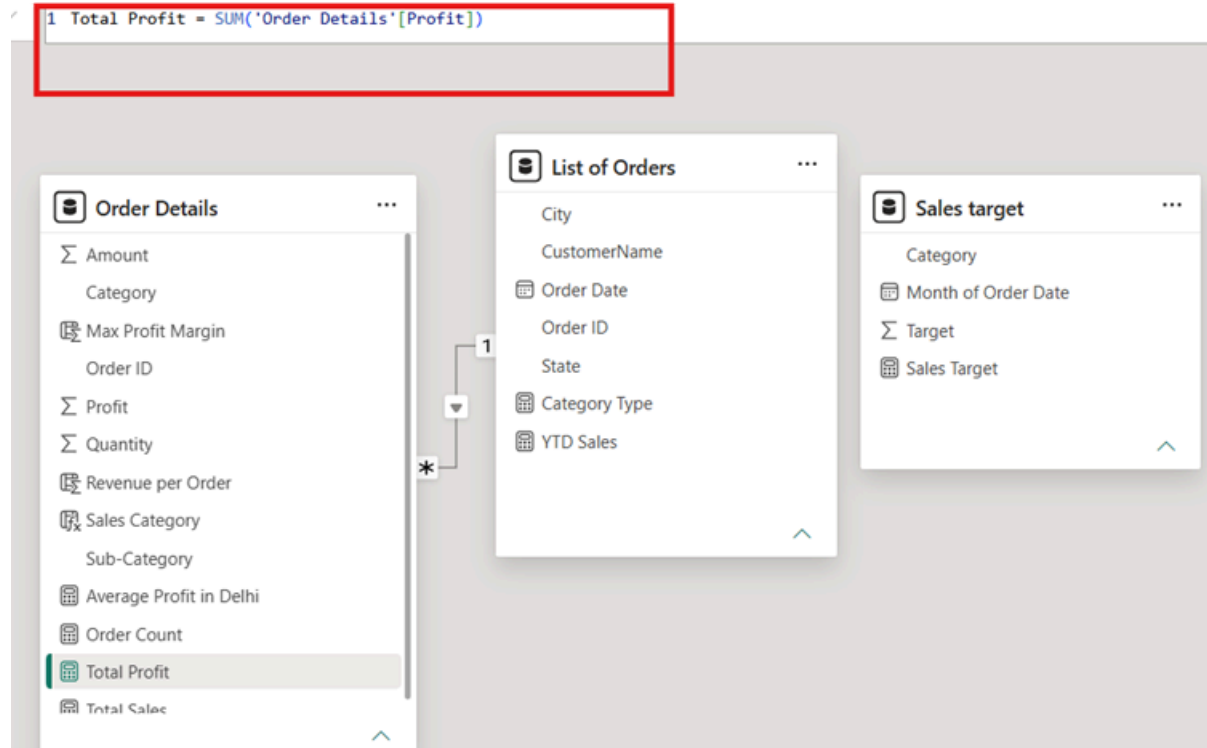
2. Max Profit Margin by Sub-Category: Analyze the maximum profit margin for each sub-category of products using a donut chart.

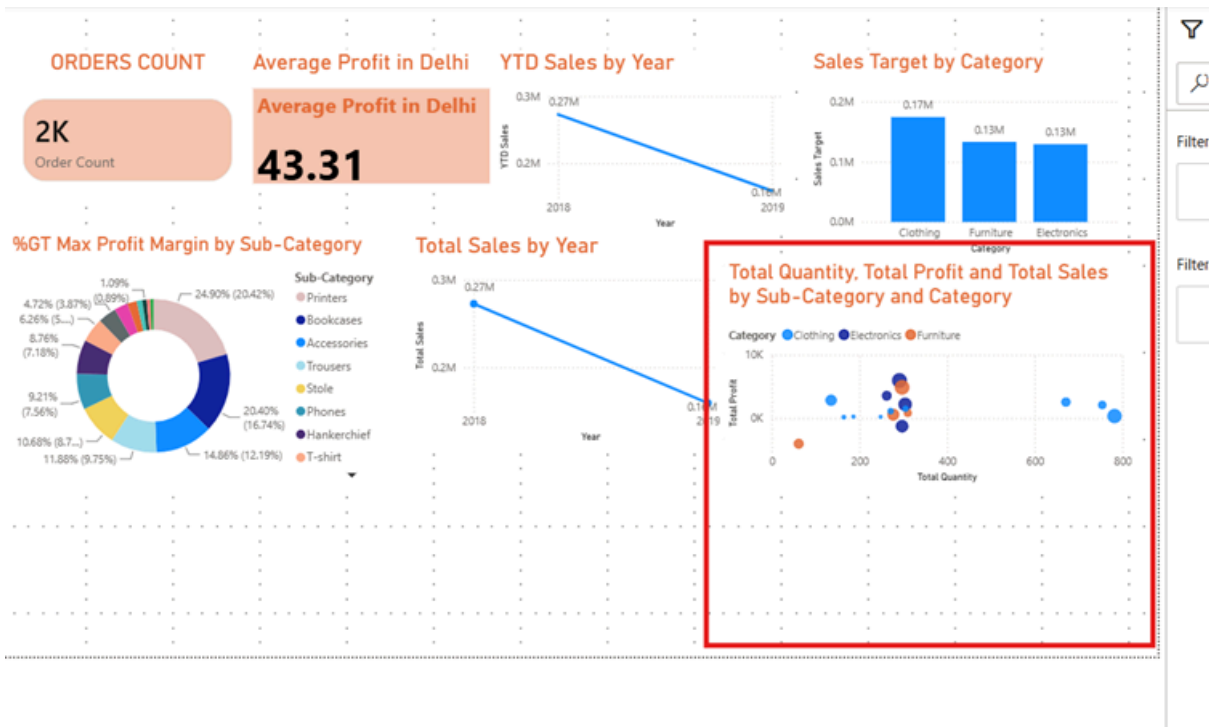
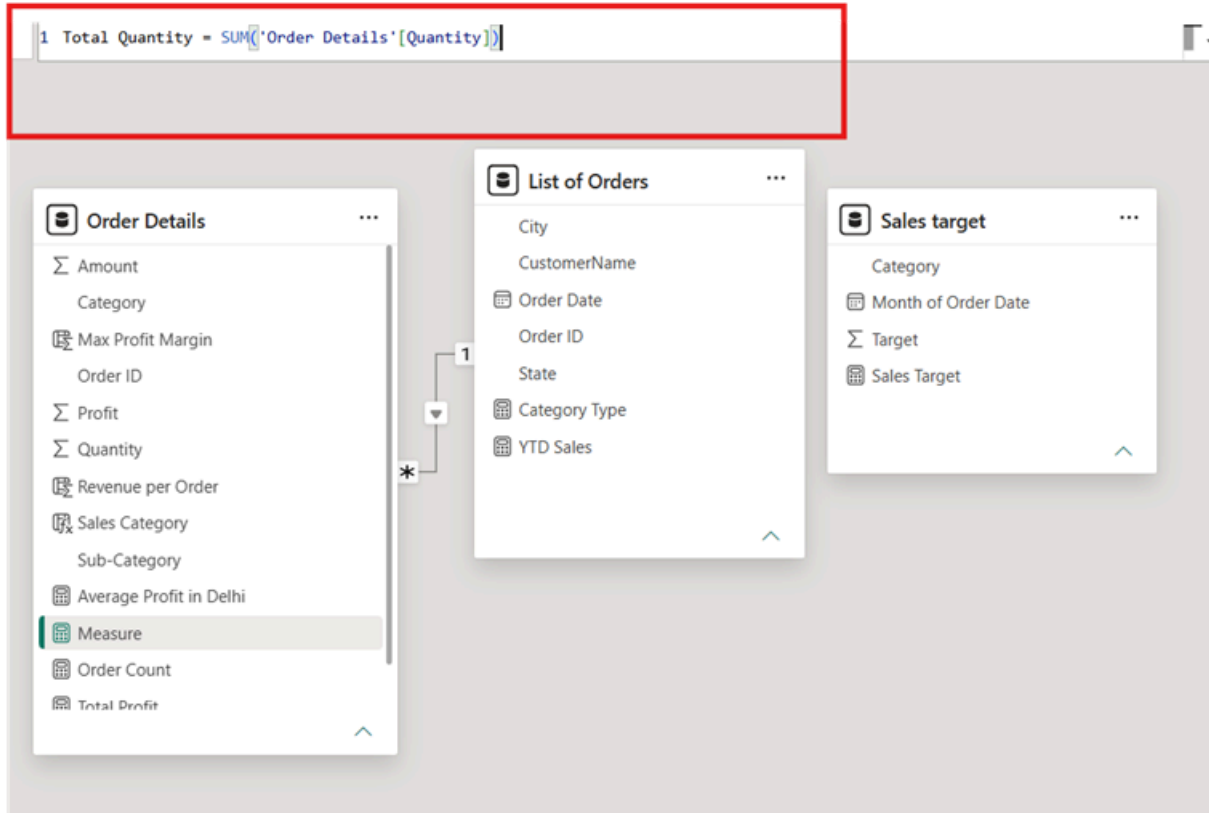


3.Monthly Sales Trend: Show the trend of monthly sales over time using a line chart.



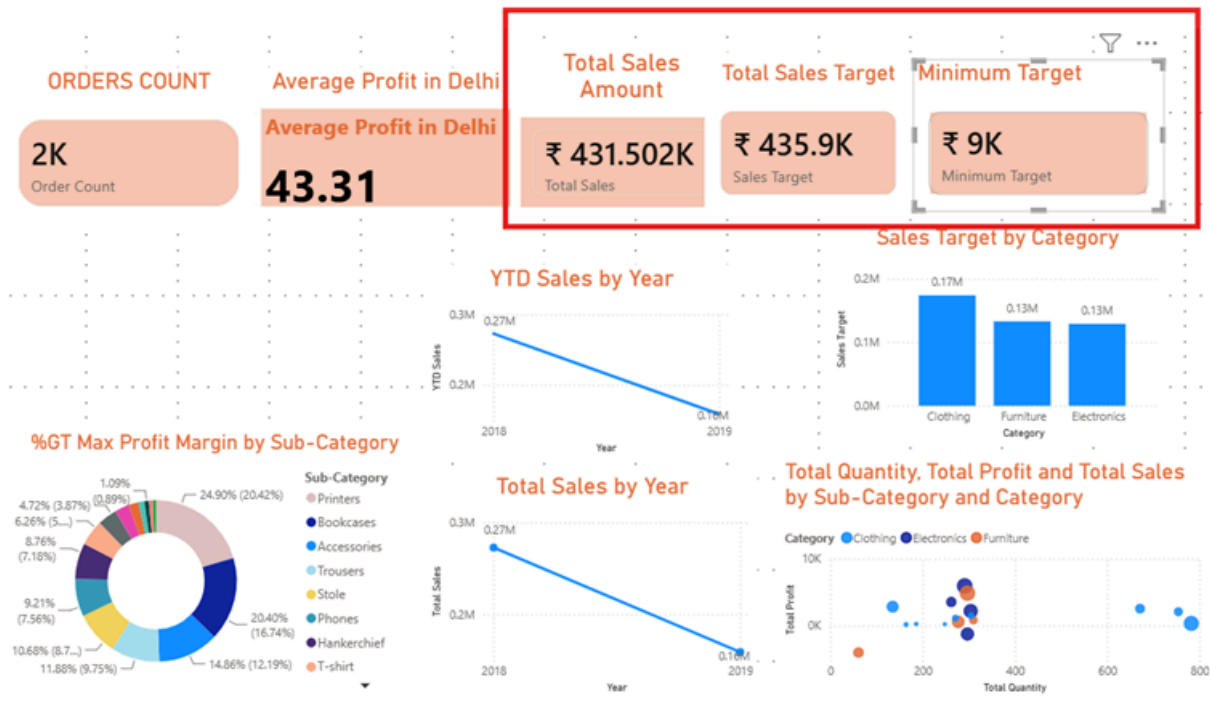
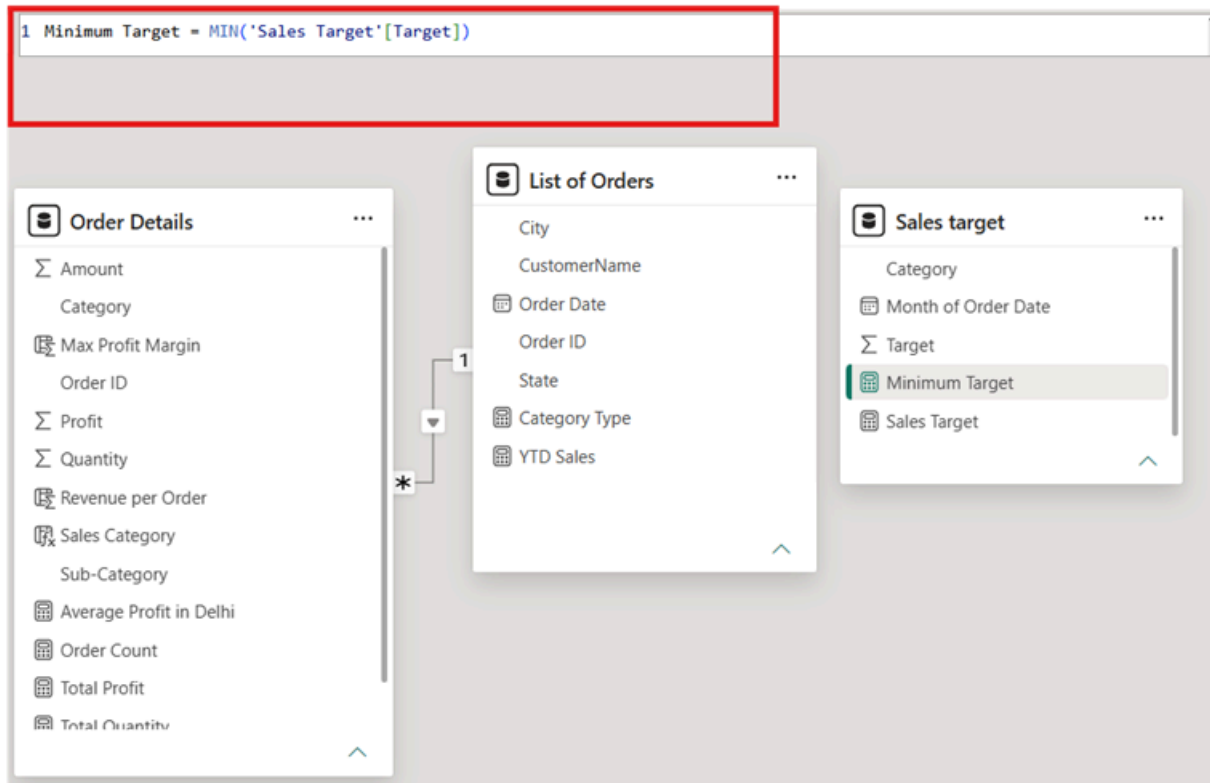
4. Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.



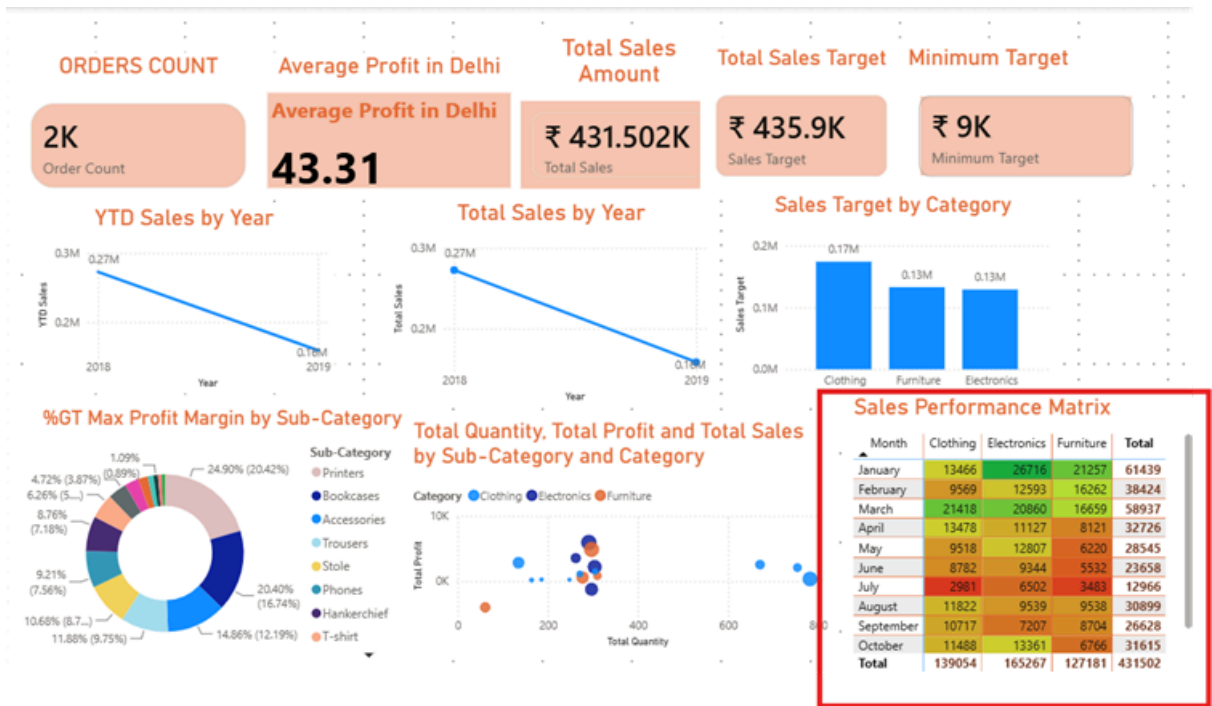


5. Comparison of Total Sales Amount and Target: Create cards to succinctly display the total sales amount alongside the sales

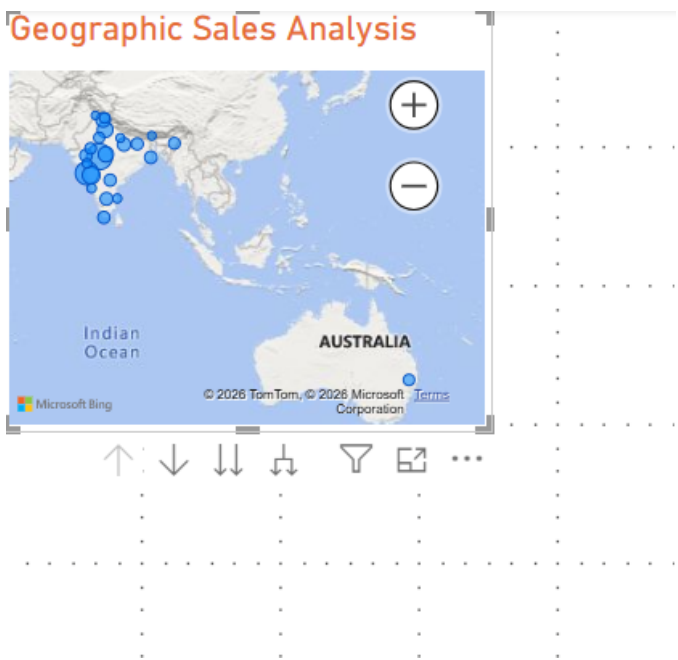
target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.



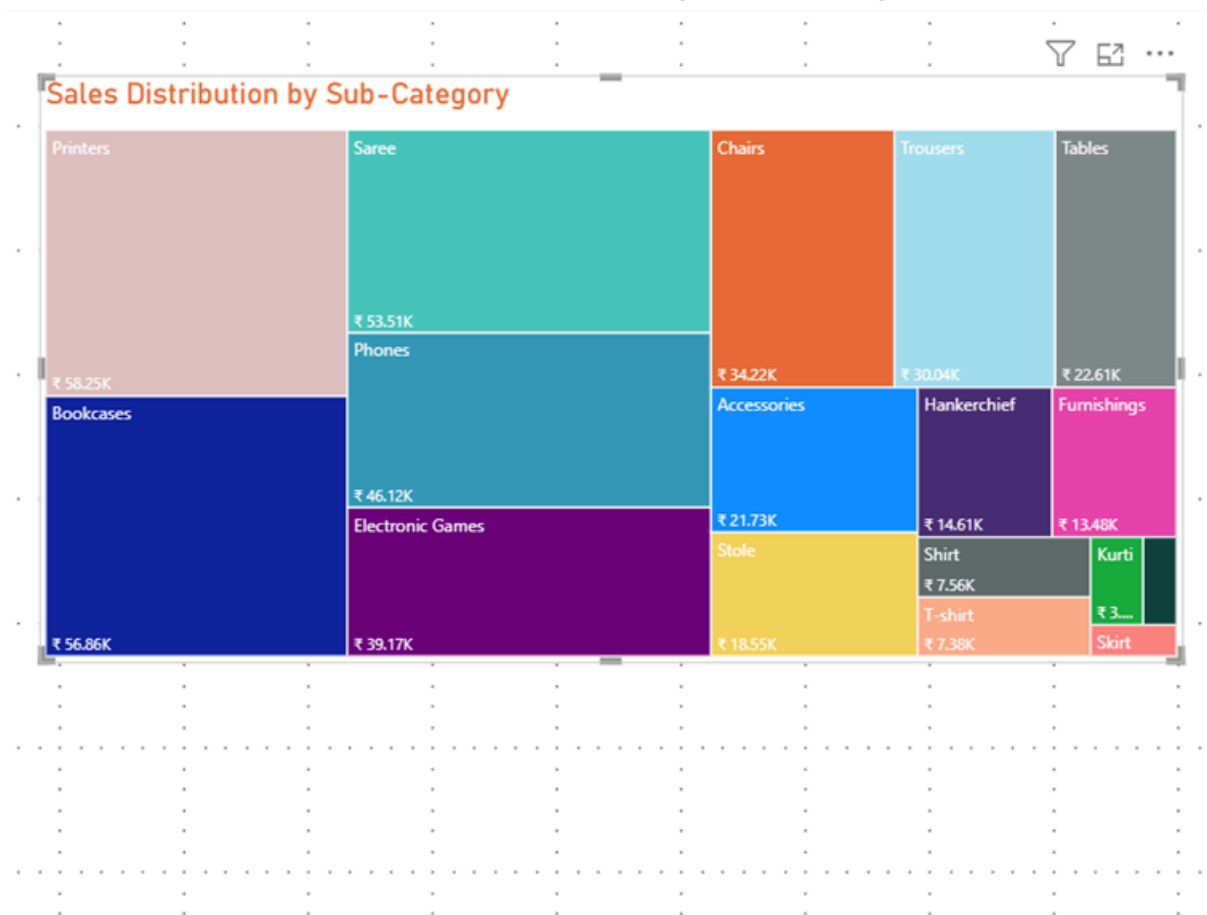
6.Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.



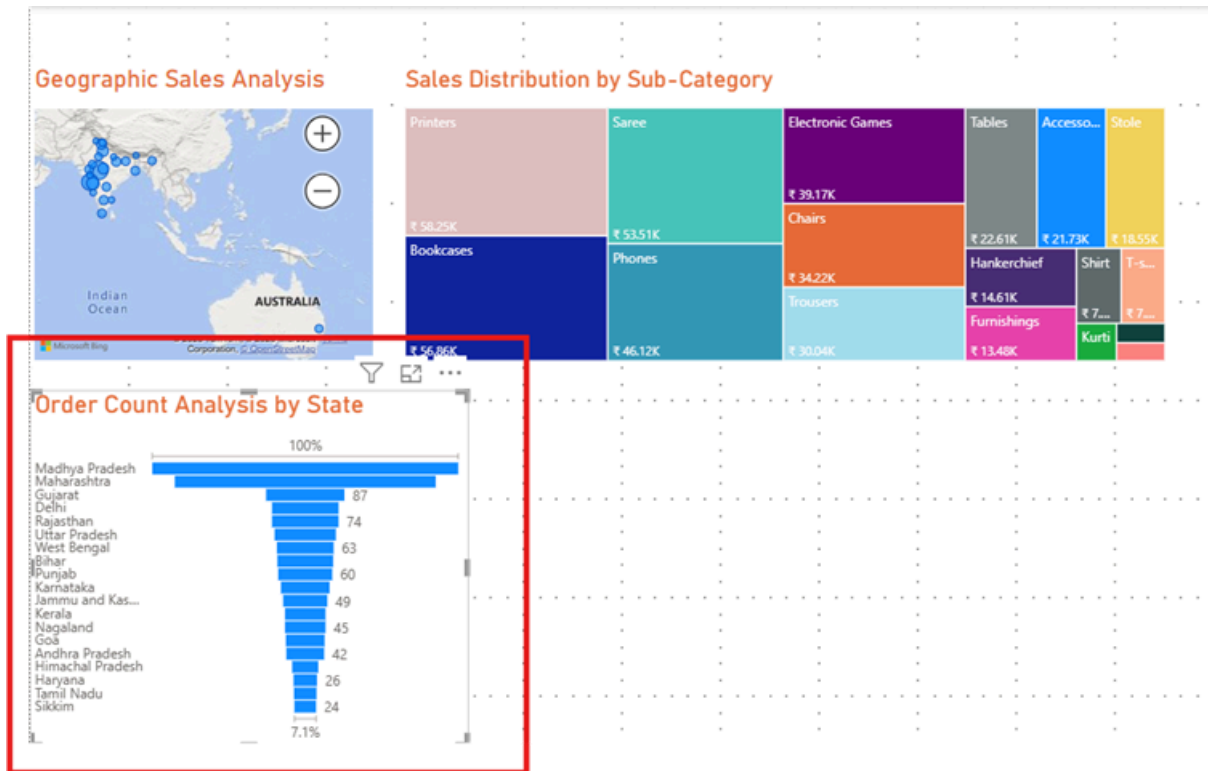
7. Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.



8. Sales Distribution by Sub-Category: Represent the sales distribution across different sub-categories using a treemap.



9. Order Count Analysis by State: Create a funnel chart to visualize the distribution of order counts across different states.



OBSERVATION:

1. The e-commerce sales dataset was successfully imported and modeled in Power BI.
2. Relationships between the tables were established to enable accurate DAX calculations and reporting.
3. A Category Type calculated column was created by combining Category and Sub-Category for improved product classification.
4. A Revenue per Order calculated column was created to calculate sales generated for each order.
5. Orders were categorized as Above Average or Below Average based on the sales amount using a calculated column.
6. DAX measures were used to calculate key business metrics such as Order Count, Average Profit, and YTD Sales.
7. The Order Count measure provides the total number of orders in the dataset.
8. The Average Profit in Delhi measure helps evaluate profitability for a specific city.

9. The Year-to-Date (YTD) Sales measure enables tracking of cumulative sales throughout the year.
10. A clustered column chart effectively compares Actual Sales with Sales Targets across product categories.
11. A donut chart highlights the distribution of Profit Margin among different product sub-categories.
12. A line chart clearly displays monthly sales trends and seasonal variations over time.
13. A scatter chart illustrates the relationship between Profit and Quantity Sold across product sub-categories.
14. KPI cards provide a quick overview of Total Sales Amount and Sales Target.
15. A matrix visual enables detailed comparison of actual sales and sales targets across categories and months.
16. A map visual identifies regional sales performance by displaying total sales for each city.
17. A Tree map effectively represents the sales contribution of different product sub-categories.
18. A funnel chart shows the distribution of order counts across different states, helping identify major sales regions.
19. Proper formatting, labels, titles, and data visualization techniques improve the readability and usability of the dashboard.
20. Overall, the dashboard provides meaningful business insights that support sales analysis, target tracking, profitability evaluation, and data-driven decision-making.